

Leong Ko Ryan Jasper

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Singapore Citizen — eligible for H-1B1 (US-Singapore FTA), a no-lottery work visa similar to TN

EDUCATION

National University of Singapore

Aug 2023 – Aug 2026

Bachelor of Engineering (Honors), Major in Computer Engineering

Singapore

- GPA: 4.88/5.0 | Dean's List; Enrolled under the Engineering Scholars Program

Stanford University

Apr 2025 – Jun 2025

Visiting Student, Spring Quarter (Exchange)

Stanford, CA

EXPERIENCE

Software Engineer

Jan 2025 – Dec 2025

ZeroshotData

San Francisco, CA

- Architected and deployed production cloud infrastructure on GCP using Terraform, GKE, Cloud SQL, and VPC, reducing environment setup time from 3 hours to 15 minutes
- Built scalable video processing apps (Python/FFmpeg/OpenCV) and ORB SLAM3 systems, reducing pipeline costs by 50% via GPU-accelerated Kubernetes autoscaling
- Built Flask API servers with Swagger/PostREST and deployed dockerized containers via Pub/Sub for parallel job processing
- Created 3D trajectory visualizer using Rerun with URDF models for interactive robotic motion analysis
- Implemented data sharding and synthetic data generation using Mosaic for scalable ML dataset augmentation

Robotics Software Engineer Intern

May 2026 – Present

DSO

Singapore

- Developed a DDS-based robot exploration simulator with LLM-powered autonomous planning agent for indoor environment navigation
- Built scene graph editor and web-based visualizer for real-time robot pose, status, and task tracking via DDS topics
- Implemented epistemic reasoning contract system enforcing evidence-backed decision making for robot action planning
- Created Python tool integration allowing LLM planner to query scene graph, compute distances, and rank exploration frontiers
- Designed prompt profiles (control/raw) and reasoning contracts for robust autonomous exploration in unknown environments

SWE Intern

May 2024 – Aug 2024

*A*STAR — National Metrology Centre*

Singapore

- Developed autonomous data-measurement web server using Python (PyVISA, Pandas, Dash) on OrangePi 3B+ for lab instrument data collection and visualization
- Engineered frequency synthesizer using Verilog and GOWIN IP cores on Tang Nano 20K, achieving 6.5–9.0 ns precision

SWE Intern

Apr 2023 – Jun 2023

HolyWally

Singapore

- Spearheaded internal tools using Python, SQL, and Superblocks with PostgreSQL, consolidating data access for ~100 users in a single dashboard
- Led QA, designing Postman calls to interface with third-party APIs (episode6) and collaborating with backend engineers to resolve UAT bugs

PROJECTS

Flair AI | *AI writing assistant; React, Tiptap, Twilio, Brainbase, Gemini, Tailwind*

2025

- **Winner (1st place, \$1,500)** — AI for Good Hackathon. Built end-to-end app with automated grading and AI voice agent for ~400 students
- Integrated **Brainbase**, **Twilio**, and **Gemini** for real-time voice feedback; built frontend with **React**, **Tiptap**, and **Tailwind CSS**

Leasy | *Lease takeover platform; Next.js 14, Mapbox GL JS, Tailwind CSS, TypeScript*

2025

- Developed Zillow-style housing platform with **Mapbox GL JS** interactive maps and advanced filtering for price/location
- Built responsive UI with **Next.js App Router** and **Tailwind CSS**, optimizing search performance for large datasets

MedBox AI | *Medical Procedure Analysis; Next.js, OpenCV, Chroma, Python*

2025

- Built intelligent surgery "black box" using **OpenCV** for real-time tool detection and **Chroma** RAG pipeline for protocol checks
- Created clinician dashboard in **Next.js** to visualize procedure timelines and automatically flag deviations

SKILLS

Full-stack: TypeScript/JavaScript, React, Electron, Tailwind CSS, Python, SQL (PostgreSQL), REST APIs, Git

Backend & Frameworks: Flask, Swagger, PostREST, Django, PostgreSQL, Node.js, Express, CI/CD, Linux, GitHub Actions

Cloud & Infrastructure: GCP (GKE, Cloud Run, Pub/Sub, Artifact Registry, IAM), Kubernetes, Terraform, Docker, Tailscale, MQTT

Data & ML: Mosaic, Rerun, ORB SLAM3, Video Processing, Synthetic Data Generation